
Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value

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PHASE 1 – BRAINSTORMING & IDEATION

1. Problem Statement:

In an organization, sensitive student and institutional information should not be accessible to every user. If all users can view, create, edit, or delete records, there is a risk of unauthorized access and modification of important data.

The project aims to implement **role-based and field-based access control** in ServiceNow so that users can access records according to their assigned roles and branch.

For example, users belonging to the **EEE branch** should be able to view EEE records, while records belonging to other branches such as ECE and CSE should not be accessible to them. Administrators should have complete access to all records.

2. Existing Problem:

The following problems can occur without proper access control:

1. Unauthorized users may view student information.
2. Users may modify records that they should not access.

3. Users may create or delete records without permission.
4. Different branches may access each other's records.
5. Sensitive institutional information may not be properly protected.
6. Managing permissions manually can become difficult as the number of users increases.

Therefore, a proper access-control mechanism is required.

3. Proposed Solution:

The proposed solution is to develop a **Script-Controlled Access Control List (ACL)** system in ServiceNow.

The system will:

1. Create an **Institution Details** custom table.
 2. Store student and institution-related information.
 3. Categorize records based on branch.
 4. Create different roles for different operations.
 5. Use ACLs to control **Read, Create, Write, and Delete** operations.
 6. Use a script to provide access based on user roles.
 7. Use a data condition to restrict records to the **EEE branch**.
 8. Provide administrators with full access.
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4. Main Idea:

The main idea of the project is:

“Allow users to access only the records and operations for which they have permission.”

The access-control system will combine:

User → Role → ACL → Branch/Data Condition → Access

For the Read operation:

- **Admin** → Can access all records.
 - **User with bb1** → Can access EEE records.
 - **User without required role** → Access denied.
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5. Objectives:

The main objectives of the project are:

1. To understand ServiceNow Access Control Lists (ACLs).
 2. To create a custom table for storing institution details.
 3. To implement role-based access control.
 4. To restrict record access based on the Branch field.
 5. To control Read, Create, Write, and Delete operations.
 6. To use server-side scripting in ACL configuration.
 7. To prevent unauthorized access to records.
 8. To provide administrators with complete access.
 9. To test the security rules using different users and roles.
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6. Proposed Features:

The project will contain the following features:

Feature	Description
Custom Table	Stores Institution Details
Branch Field	Identifies EEE, ECE, and CSE records
Role-Based Access	Controls user permissions
Read ACL	Controls who can view records
Create ACL	Controls who can create records
Write ACL	Controls who can edit records
Delete ACL	Controls who can delete records
Script Control	Validates user roles
Admin Access	Provides full access to administrators

7. Expected Outcome:

After implementing the project:

1. EEE users will be able to access permitted EEE records.
 2. Unauthorized users will not be able to access restricted records.
 3. Users will only get Create, Write, or Delete permissions according to their roles.
 4. Administrators will have full access.
 5. The system will provide better security for institutional information.
 6. The project will demonstrate how ServiceNow ACLs and scripts can be used for customized access control.
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8. Conclusion:

The brainstorming phase identified the need for a secure and flexible access-control mechanism in ServiceNow. The proposed **Script-Controlled ACL** solution addresses unauthorized record access by combining **roles, ACLs, scripts, and branch-based data conditions**.

This idea will be developed further in the next phases, including **Requirement Analysis, Project Design, Planning, Development, Testing, Documentation, and Demonstration**.
